COGNITIVE FUNCTION & NEUROPROTECTION

Semax — the full review

ACTH-DERIVED SYNTHETIC PEPTIDE

This is the deeper look behind the short version — what the Russian stroke-recovery trials actually show, why that is a different claim than cognitive enhancement, common misunderstandings, and what your consultation covers.

What it is

Semax is a lab-made chain of seven amino acids copied from a small fragment of ACTH (adrenocorticotropic hormone), the hormone the pituitary gland normally releases to trigger cortisol production. It was modified to retain some of ACTH's effects on the brain and nervous system while losing ACTH's hormonal effects on the adrenal glands, so it is not expected to raise cortisol the way natural ACTH does.

In Russia, Semax is a licensed prescription nasal-drop medicine, approved in 2011 and placed on Russia's List of Vital & Essential Drugs, used mainly for people recovering from ischemic stroke and for certain other neurological conditions. In the US wellness market it is sold very differently — almost always as a nasal spray, sometimes for injection — and marketed as a nootropic for focus, memory, and mental performance in otherwise healthy people. That US marketing use is not the use it was studied or approved for in Russia, and it has never been evaluated by FDA for any purpose.

Fewer than 50 healthy adults have ever been studied taking Semax, in short dose-finding studies that did not measure focus or memory as an outcome — a meaningful gap between the stroke-recovery evidence that exists and the cognitive-enhancement claim that drives most current sales.

What people use it for

USED FOR	EVIDENCE	WHAT THAT MEANS
Recovery after ischemic stroke (the Russian-approved use)	**Human trials**	Real trials exist since the 1990s, but they are small and have never been replicated outside Russia.
Cognitive focus or memory in healthy adults	**Limited human data**	Fewer than 50 healthy adults have been studied, in studies that did not measure cognition.
Memory or focus enhancement specifically	**Limited human data**	No identified human trial measured memory or focus as an outcome in healthy people; the mechanistic story is drawn from rat studies.
ADHD-like focus or attention claims	**Theoretical**	No human or animal study of Semax for attention specifically was identified; this appears to be an inference from its dopaminergic mechanism.
General neuroprotection beyond stroke	**Animal data only**	Supported by rodent studies in stroke models; no human data exists outside the stroke trials.

How it works

The proposed mechanism centers on raising BDNF and related neurotrophic signals, affecting serotonin and dopamine signaling systems, and shifting which genes get switched on in brain tissue after injury in animal stroke models. FDA's 2026 review describes Semax's mechanism as working through "poorly understood mechanisms of action."

A notable animal finding deserves mention: Semax potentiated amphetamine-induced dopamine release and locomotor activity in mice. FDA flagged this as concerning, since increased dopaminergic tone in this brain region is a response typically induced by drugs of abuse, and no study has tested whether Semax itself has reinforcing or addictive properties. One small Russian human study examined BDNF levels alongside motor recovery in stroke patients — the closest thing to human mechanistic confirmation, though it is a single, small trial.

What the research actually looked at

FDA's own 2026 review counts roughly 550 documented human exposures across Semax's entire history: 33 to 47 healthy adults in short dose-finding pharmacology studies that did not measure cognitive performance, 69 adults with diagnosed neurological conditions such as chronic ischemic brain disease, migraine, and trigeminal neuralgia, and 451 children treated in Russian clinical settings for depression or tics — by far the largest population studied, and one with little bearing on the healthy-adult cognitive-enhancement claim Semax is sold on in the US.

Two claims need to be kept separate, because marketing blurs them. Post-stroke neuroprotection and recovery — the actual Russian-approved use — rests on real human trials in a real patient population spanning decades, though limited to a single Russian research lineage, small samples, and no independent Western replication. Cognitive enhancement in healthy people — the claim actually driving US sales — has essentially no dedicated evidence; controlled trials for this specific use in healthy adults essentially do not exist.

Tellingly, when FDA's own reviewers examined largely this same body of Russian clinical evidence in 2026, they concluded it showed "insufficient evidence of effectiveness" — even for the narrower, disease-specific indications actually nominated, a materially lower bar than the healthy-adult cognitive-enhancement claim driving US sales.

Common misunderstandings

- **“It's an approved smart drug.”** Semax is a government-approved prescription drug in Russia — but for ischemic stroke and related neurological conditions in a diagnosed patient population, not for cognitive enhancement in healthy people, and not by FDA or any Western regulator.
- **“The stroke research proves it works as a nootropic.”** These are not the same claim. The stroke-recovery evidence is about restoring function after a diagnosed brain injury. It says nothing about whether an otherwise-healthy adult will think faster, remember more, or focus better.
- **“It has no side effects.”** FDA's adverse-event system logged a direct consumer report of ocular pain and eye burning after intranasal Semax purchased online, with hospitalization and symptoms still unresolved a year later; nasal irritation with chronic use is commonly reported, and whether Semax has abuse potential has never been formally studied.
- **“It's just a natural hormone fragment, so it's basically safe.”** Semax is a synthetic, lab-modified peptide engineered specifically to strip out ACTH's normal hormonal activity while keeping other central-nervous-system effects — it is not something the body produces on its own, and FDA additionally found it "not well-characterized" as a compounding ingredient.

Who it is not for

- Pregnancy or breastfeeding, or trying to conceive — no reproductive or developmental toxicity data were identified.
- Active psychosis or significant psychiatric instability, given Semax's direct effects on dopaminergic and serotonergic signaling.
- A personal or family history of substance-use disorder, or stimulant sensitivity — Semax potentiates amphetamine-induced dopamine release in animal studies, and no abuse-potential studies exist.
- Significant nasal mucosal disease or recent nasal surgery — nasal irritation is the most consistently reported effect of the intranasal form.
- Subcutaneous injection use — every published human study used the intranasal route; no human data exists for injection.
- Substituting for emergency stroke care.
- Competitive athletes under WADA Code without first checking current status.

Competitive athletes: Semax is not explicitly named on the WADA Prohibited List. Because it does have current government approval — Russia's — its fit under WADA's S0 ("Non-Approved Substances") catch-all is genuinely unresolved rather than settled, and no confirmed sanction case specific to Semax was located. Status should be treated as unresolved and high-risk, not cleared, and a competing athlete should check Global DRO or their national anti-doping organization before use.

Who makes this, and under what rules

Everything dispensed here is prepared by a licensed 503A compounding pharmacy — not bought from a supplier and repackaged, and not sourced from an online seller.

That designation carries specific, enforceable obligations:

- **A prescription in your name.** Compounded from a prescription for one identified patient, after an evaluation.
- **Licensed and inspected.** A state licence, subject to inspection by the state board of pharmacy.
- **Ingredients that can be traced.** A certificate of analysis from an FDA-registered facility, back to a named source and lot.
- **Sterile preparation to a published standard.** USP standards.
- **A paper trail.** A lot number and beyond-use date on every preparation.

A compounded preparation is not an FDA-approved medicine — but it is the difference between a licensed pharmacy working to a standard and an unregulated website.

Where this stands

Approved in Russia, for a different use than it's sold for here. A licensed stroke-recovery medicine there, never reviewed by FDA. The July 2026 advisory committee separately voted 8-5 to recommend it for a compounding list, for cerebral ischemia, migraine, and trigeminal neuralgia — not for the cognitive-enhancement claim driving US sales.

Fewer than 50 healthy adults have ever been studied. The human evidence base overwhelmingly concerns stroke patients, not people seeking a cognitive edge.

The stroke-recovery evidence and the smart-drug marketing are not the same claim, and only one of them has been tested.

Clinical detail

Semax was first described in the Russian scientific literature in 1991 and formally approved by the Russian government on December 7, 2011.

It was re-nominated and formally reviewed at the July 23–24, 2026 FDA Pharmacy Compounding Advisory Committee meeting, for cerebral ischemia, migraine, and trigeminal neuralgia — not for cognitive enhancement; FDA staff recommended against inclusion, and the committee voted 8-5, with one abstention, to recommend it anyway. The vote is non-binding and has not been adopted.

FDA's adverse-event system logged a direct consumer report of ocular pain and eye burning after intranasal Semax purchased online, with hospitalization and unresolved symptoms a year later.

The largest identified Russian stroke trial enrolled just 30 patients in the acute phase of hemispheric ischemic stroke.

The information on this page is general education. It is not medical advice and it is not a recommendation for any treatment.

Semax is not an FDA-approved medicine for the uses described here. A compounded preparation is not an FDA-approved product and has not been reviewed by the FDA for safety, effectiveness or quality.

Descriptions of what this is used for reflect how it is commonly used and discussed. They are not claims that it is effective for any purpose.

Any prescription follows an individual evaluation by a licensed provider and is issued only where that provider judges it clinically appropriate. A prescription is never guaranteed.